



CA Manager

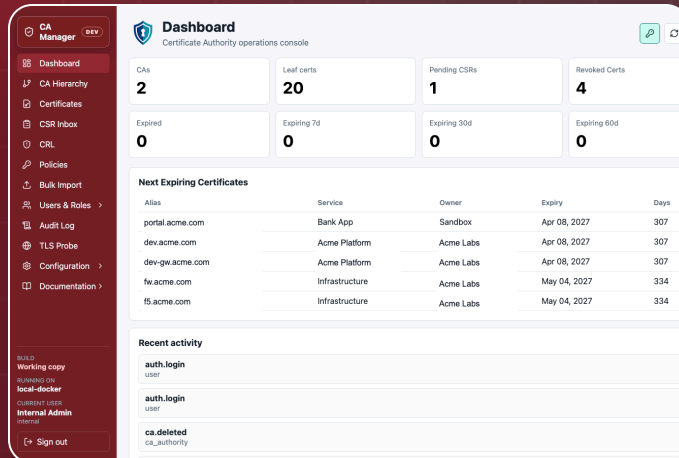
PRIVATE PKI OPERATIONS CONSOLE

FEATURE AND TECHNOLOGY BRIEF

BUILT FOR LAB PKI TEAMS

Private certificate operations without losing the thread.

CA Manager is a browser-based control plane for internal, lightweight certificate authority management: CA hierarchy, leaf certificates, CSRs, revocation, policy, identity, automation, and TLS inspection in one place.



PKI

Create, import, retire, revoke, renew, and export private PKI assets.

RBAC

Roles, teams, LDAP-backed users, service accounts, and audit trails.

ACME

Internal policy-bound enrollment with certbot, lego, and acme.sh compatibility.

SCEP

One-time challenge enrollment for devices and managed clients.

PRODUCT SCOPE

Useful where manual OpenSSL stops scaling.

Best for development stacks, sandboxes, internal platform experiments, training PKI, homelabs, and non-critical private apps that need repeatable certificate workflows without public-trust CA overhead.

01

Operate the hierarchy

Root and intermediate CAs stay visible with status, ownership, issuer relationships, details, and lifecycle controls.

02

Manage inventory

Track leaf certificates, SANs, services, environments, expiry windows, revocation state, and stored private keys.

03

Support automation

Issue scoped API keys, expose ACME/SCEP paths, publish CRL/OCSP endpoints, and inspect live TLS endpoints.



PKI Operations

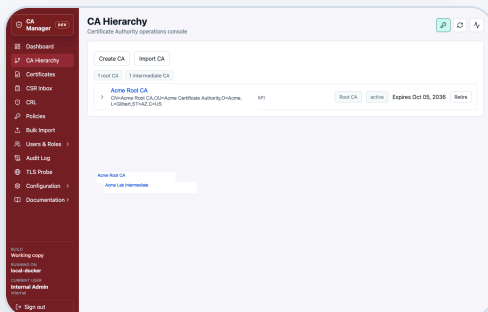
CAS, CERTIFICATES, CSRS, AND REVOCATION

FEATURE DEEP DIVE

DAILY WORKFLOWS

Move common CA work into a visible operating model.

The UI is organized around the real objects lab teams touch: CA hierarchy, certificate inventory, signing requests, revocation lists, policies, and operational audit history.



CA lifecycle

Create root CAs, create intermediate CAs from active roots, import existing CAs, retire authorities, and delete only when active child dependencies allow it.

Key custody

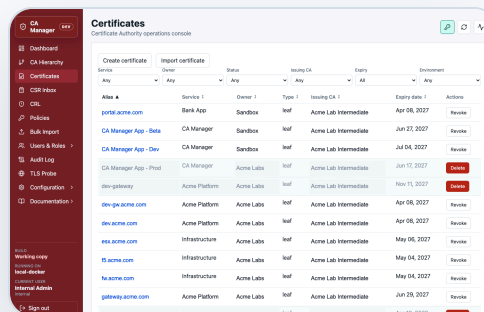
Private keys are encrypted before storage. CA and leaf private-key export can be controlled by permission and optional security-key step-up.

Certificate inventory

Track common names, SANs, status, issuer, owner, service, environment, expiry, chain metadata, and export formats.

CSR intake

Import CSRs, review request details, sign through policy-bound CA workflows, or reject requests with a recorded reason.



CRL publication

Serve DER and PEM CRLs from public endpoints and cache signed CRLs per CA.

OCSP responder

Expose DER OCSP request/response paths for revocation status checks.

Policy extensions

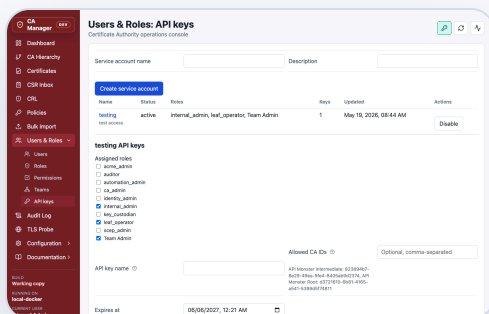
Embed CRL distribution and OCSP AIA URLs in issued certificates using policy placeholders.



AUTOMATION SURFACE

Scriptable paths for lab services and device enrollment.

CA Manager supports browser-led operations and non-human workflows. Automation identities use scoped API keys; protocol clients can enroll through internal policy-bound ACME and SCEP surfaces.



SURFACE

SUPPORTED CAPABILITY

Service accounts

Non-human identities for CI/CD jobs, scripts, and certificate automation.

API keys

X-CA-Manager-API-Key authentication, role-derived permissions, expiry, status, and optional CA-ID scoping.

Audit events

Operational events recorded for login, issuance, revocation, automation, config, and protocol activity.

A

ACME

RFC 8555-style directory, accounts, orders, authorizations, challenges, replay nonces, external account binding, revocation, and account key rollover for internal PKI automation.

S

SCEP

Policy-bound enrollment endpoints with GetCACaps, GetCACert, PKIOperation, one-time challenge passwords, DNS suffix checks, and maximum validity controls.

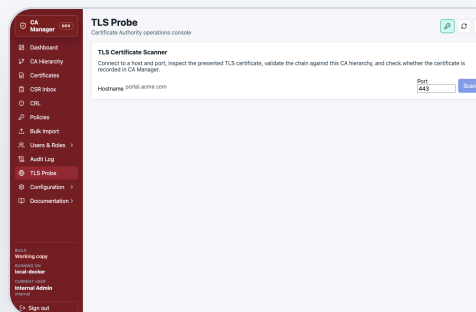
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TLS Probe

Connect to a host and port, retrieve the presented TLS chain, validate against managed CAs, and identify whether the leaf certificate is recorded.

Client compatibility

- > ACME compatibility has been verified with certbot, lego, and acme.sh against the local ACME directory.
- > ACME challenge handling is modeled as internal policy validation, not public domain-control validation.
- > SCEP is intended for device and managed-client workflows that can supply policy-issued challenge passwords.



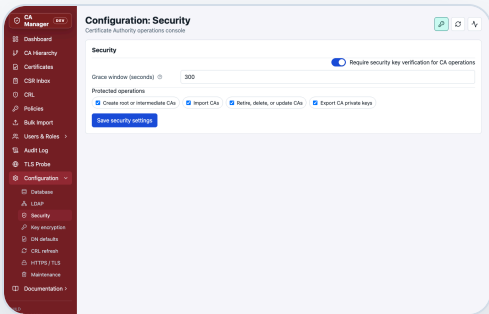


HOW IT RUNS

Simple deployment pieces with explicit trust boundaries.

The stack is built around Docker Compose, a Node API, a web UI, Nginx as the published endpoint, MySQL as durable storage, and externally supplied secrets for session and private-key encryption.

AREA	SUPPORTED TECHNOLOGY AND BEHAVIOR
Runtime	Docker Compose app containers. Nginx serves the web UI and proxies /api , /acme , /scep , /crl , /ocsp , and health endpoints.
Database	MySQL stores CA records, certificates, encrypted private keys, users, roles, audit events, protocol data, and configuration.
Secrets	KEY_ENCRYPTION_SECRET stays outside MySQL and encrypts stored private keys. SESSION_SECRET protects browser sessions.
Identity	Internal users, imported LDAP-backed users, OpenLDAP and Active Directory provider settings, roles, teams, and group-role mappings.
Security keys	Optional WebAuthn/FIDO2 step-up for protected CA operations with a configurable grace window.
HTTPS	Certificate selection in the UI writes cert/key material to the Nginx SSL volume; Nginx reloads automatically after changes.



Access model

Built-in and custom roles cover CA administration, leaf operations, audit, automation, ACME, key custody, SCEP, and identity administration.

Maintenance model

Clear test-data categories while preserving database, LDAP, key-encryption, HTTPS/TLS, permissions, and system-role configuration.

Best fit

- > Development and QA infrastructure
- > Homelab and private service stacks
- > Internal platform experiments
- > Training, demos, and disposable PKI

Not the target

- > Public-trust certificate authority operations
- > Bank-grade or regulated production PKI without additional controls
- > Environments where losing MySQL or KEY_ENCRYPTION_SECRET is unacceptable without tested backups